

VICTOR ANTONIO CRUZ

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EDUCATION

University of California, Santa Cruz

BS Computer Science

Santa Cruz, CA

Sep 2024 - June 2028

Relevant Classwork: Programming Abstractions; Python, Computer Systems and Assembly, Introduction to Data Structures and Algorithms Language, Discrete Mathematics, Introduction to Machine Learning

PROJECTS

California Housing Price Predictor

- Performed exploratory data analysis on the 20,000-entry 1990 California Census housing dataset, identifying key relationships between features and handling missing values.
- Preprocessed data by scaling numeric features, encoding categorical variables, and engineering new features to improve model inputs.
- Trained and compared Linear Regression and Decision Tree models in Python/Scikit-learn, evaluating performance with R^2 and MSE using proper train/test splits and cross-validation.
- Visualized predictions and feature importance to communicate which factors most influenced housing prices, demonstrating end-to-end analysis and interpretation.

EXPERIENCE

Undergraduate Research Auditor — AEAI Lab

University of California, Santa Cruz

Santa Cruz, CA

2025 Sep- 2025 Dec

- Onboarded to research infrastructure by learning Docker, Kubernetes, and Nautilus, then deployed and ran experiments in a reproducible environment.
- Set up and used the CARLA simulator (CARLA Gym) to run autonomous-driving reinforcement learning experiments under a discrete action-space control setup.
- Read and implemented reinforcement learning papers, including Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC), and documented design choices for training stability.
- Built a benchmarking framework to run PPO and discrete SAC end-to-end and generate comparison plots (episode return, episode length, entropy) to analyze learning behavior.

Undergraduate Research Assistant – VORTREX Computational Modeling

University of California, Santa Cruz

Santa Cruz, CA

2026 Jan – Present

- Developing MATLAB simulations using matrix-based formulations to model force interactions and motion dynamics.
- Currently implementing Stokeslet methods to capture viscous flow interactions in realistic environments.
- Collaborate with faculty mentor to iteratively refine mathematical models and validate simulation behavior.

SKILLS

Programming Language:	Python, C, C++, MATLAB, Assembly Language, R
ML/DL Libraries:	PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy
RL Libraries:	Stable-Baselines3, OpenAI Gym, CARLA Gym
Visualization:	Matplotlib, Seaborn, TensorBoard
DevOps & Systems:	Docker, Kubernetes, Nautilus

ACTIVITIES

Baskin Day 2025 – Internal Member: Collaborated with the internal team to organize and execute Baskin Day, facilitating meetings with clubs, coordinating participation, and working with faculty.

CruzHacks 2025 – Sponsorship Lead: Secured and managed sponsorships independently, coordinating with partners to deliver sponsor benefits and support the hackathon's mission of innovation and collaboration.